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Editorial Board,
Journal of Nuclear Materials

Dear Editors,

On behalf of my co-authors and myself, I am hereby submitting a revised version of our manuscript, entitled “Xe gas bubble re-resolution in U-10Mo nuclear fuel”, for publication in the Journal of Nuclear Materials. A previous submission of the manuscript (JNUMA-D-26-00476) was not compliant with the journal’s preferred article structure. The revised version fixes the problem by providing both Results and Discussion sections.

This study addresses a critical challenge in the evaluation of U-10Mo nuclear fuel: predicting the swelling behavior during the lifetime of the fuel. A key driver of this swelling is the evolution of Xe gas bubbles, which is governed by the complex interplay between gas atom trapping and radiation-induced “re-resolution,” the process where fission fragments knock gas atoms back into the fuel matrix.

Our research explored all potential re-resolution mechanisms, including both collision cascade-driven homogeneous re-resolution and thermal spike-driven heterogeneous re-resolution. To this end, we employed molecular dynamics simulations coupled with the two-temperature model, and binary collision approximation (BCA) simulations. Because existing BCA tools lacked the capability to model inhomogeneous 3D systems, we developed a custom geometry within the *RustBCA* code to simulate gas bubbles embedded in a solid fuel matrix. Our research resulted in the quantification of the re-resolution rate in U-10Mo as a function of bubble size, bubble pressure, and fission rate density. To our knowledge, this is the most extensive investigation of the re-resolution behavior in any nuclear fuel to date. This work is highly appropriate for Journal of Nuclear Materials as it presents a novel computational approach for the quantification of re-resolution in nuclear fuels.

We believe our research offers valuable insights into the behavior of the U-10Mo fuel and adds greatly to the scientific community.

Sincerely,

Benjamin Beeler